



IT Skills



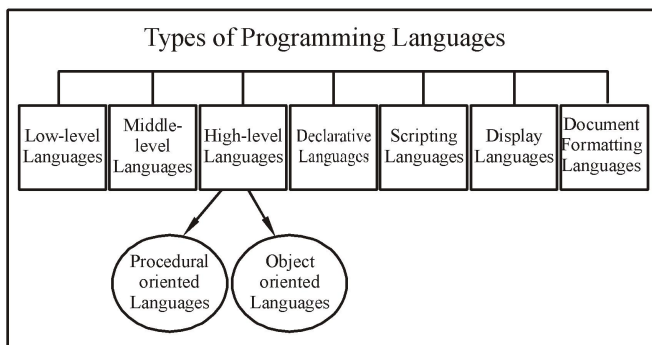
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Introduction to Basics of Coding

1.1 INTRODUCTION TO COMPUTER PROGRAMMING

A computer Program is sequential set of instruction written in same computer language that is used to direct the computer to perform some specific task of computation.

Different types of Programming Languages



Low-Level Programming Languages

Low-level programming languages again consists of machine-level languages and assembly-level languages.

- **Machine Language** - Machine language comes under the category of low-level programming language that is built up of 0's and 1's. Some high-level languages are compiled into machine-level languages, so the computer understands the code.
- **Assembly Language** - Assembly language also comes under the category of low-level programming language that is compiled by an assembler. Translation of human-written code is converted into machine code by these assemblers only.

Middle-Level Programming Languages

Middle-level programming languages are the programming languages that have characteristics of both low-level and high-level languages.

C Language - The c programming language acts as a mediator between hardware and programming layer. C is easy to understand and flexible. It is a compiled language and does not use objects and classes. Compilers and editors use C to design their applications.

High-Level Programming Languages

Procedural Languages - Procedural languages are also programming languages in which the written code will pass through many procedures, and the computer executes the program

Examples of procedural languages are **FORTRAN, COBOL, SQL, and GO.**

Object-Oriented languages - The object-oriented programming languages which use objects and classes to write code. The usage of objects and classes makes it to solve real-world problems.

object-oriented programming languages are **Java, R, Ruby, Python, C#, JavaScript and Perl.**

Declarative Languages

A declarative language is a programming language where the programmers concentrate on the goal or result of the program.

An example for a declarative language is "PROLOG," which is similar to SQL

1.2 ALGORITHMS

The word Algorithm means "a process or set of rules to be followed in calculations or other problem-solving operations". Therefore Algorithm refers to a set of rules/ instructions that step-by-step define how a work is to be executed upon in order to get the expected results.

An algorithm is defined as step by step procedure to solve a given problem by accepting set of inputs and producing the desired output.

Business Process Automation/ERP

3.1 INTRODUCTION TO BUSINESS PROCESS AUTOMATION:

Business Process Automation also known as business automation or digital transformation is the technology enabled automation of complex business process. It is use of technology to execute difficult tasks or processes in a business manual effort can be replaced.

Advantages:

1. Improve Accountability, Transparency and enable accurate data.
2. Increase efficiency.
3. Increases Productivity.
4. Reduced operating costs.
5. Reduced time in task execution

3.2 Organization structure:

An organization is a social unit of individual that is designed and managed to achieve collective goals.

An organizational structure is a visual diagram of a company that describes what employee's do, when they report to and how decisions are made across the business.

The process for creating an organizational structure

Plan the future

Consider the past

Build your organizational structure

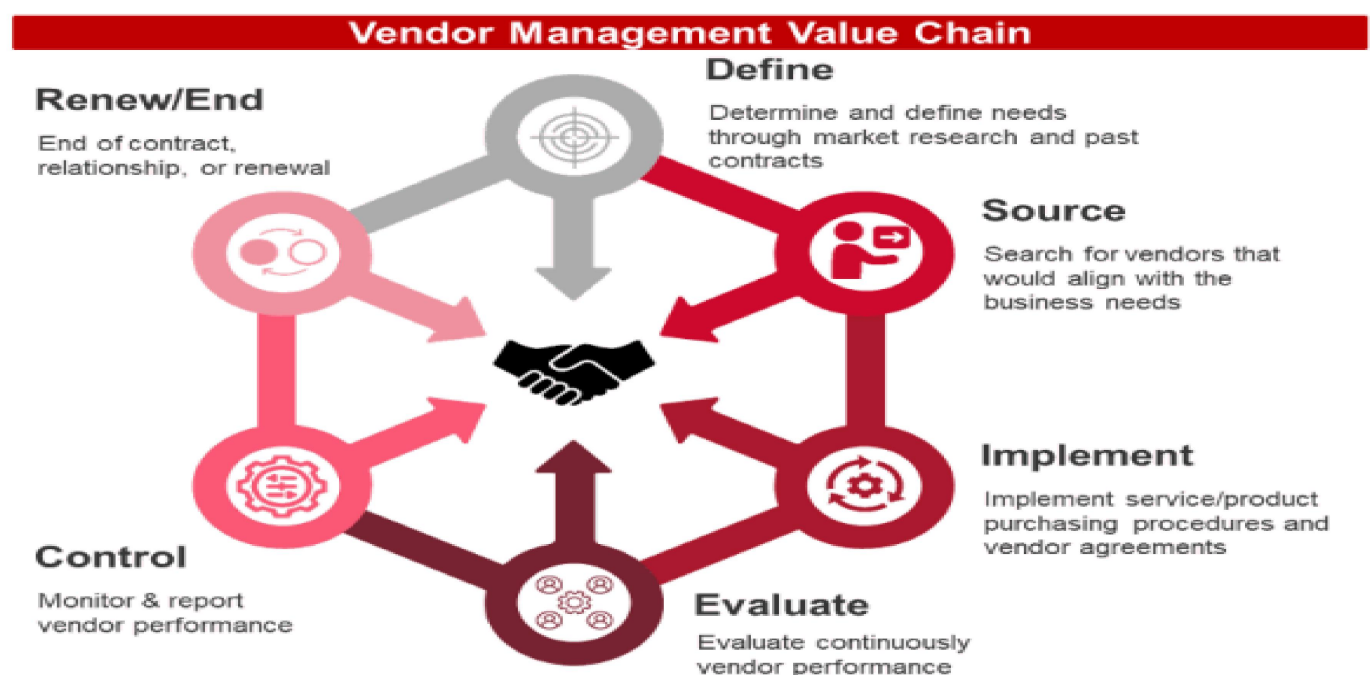
Fill in the people

Balance authority and responsibility

Fill in employee data and metrics

Practice robust performance management of employees

Review your organizational structure annually



Introduction to Cloud and IOT Concepts

4.1 FUNDAMENTALS OF CLOUD:

Cloud computing is a subscription based service where can obtain networked storage space and computer resource over the internet.

Characteristics of cloud Computing:

- Pay as you use
- No server space required.
- No experts required for hardware and software maintenance.
- Better Data security
- High Flexibility
- Data can be accessed and shared anywhere over the internet.
- Data Recovery is easy.

Cloud computing divided into two

1. Deployment Model
2. Service Model

Deployment Model Types:

- **Public Cloud:** A public can be accessed by any subscriber with an internet connection and access to the cloud space.
Ex: Microsoft Azure, Google, AWS (Amazon web service)
- **Private Cloud:** A private cloud is established for a specific group or organization and limits access to just that group.
Ex: VMware
- **Hybrid Cloud:** A hybrid cloud is a combination of at least two clouds where the clouds include either mixture of public and private.
- **Community Cloud:** A community cloud is shared among two or more organization that have similar requirement.

4.2 SERVICE MODEL TYPES

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)

- **Infrastructure as a Service (IaaS):** IaaS is also known as Hardware as a Service (HaaS). It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS:

There are the following characteristics of IaaS -

- Resources are available as a service
- Services are highly scalable
- Dynamic and flexible
- GUI and API-based access
- Automated administrative tasks

Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud

- **Platform as a Service (PaaS):** PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

Characteristics of PaaS:

There are the following characteristics of PaaS -

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- Support multiple languages and frameworks.
- Provides ability to “**Auto-scale**”.

Cyber security And Safety

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5.1 INTRODUCTION TO CYBER SECURITY AND SAFETY

Cyber Security

Cyber Security is the technology and process that is designed to protect networks and device for attacks, damage or unauthorized access.

Cyber Safety

Cyber Safety is the safe and responsible use of information and communication technology. It is about helping information safe and secure.

5.1.1 Brief Awareness on cyber Safety Measures

- Keep personal information professional and limited
- Keeps the privacy setting on.
- Practice safe browsing.
- Make sure your internet connection is secure.
- Be careful what you download.
- Choose strong Passwords
- Make online purchases from secure sites.
- Keep your Antivirus program up to date
- Block third party website.

5.1.2 Basic Security Issues in Mobile phones

1. **Data Leakage:** Mobile application is often the course of unintentional data leakage.
2. **Unsecured Wi-Fi:** Free wifi networks are usually unsecured. Never use it to access confidential or personal services like banking and credit card information.
3. **Spyware:** It is unwanted software designed to access your mobile often without your knowledge.
4. **Unsafe Operating System:** Large number of mobile devices is not kept up to date with operating system release.

5. **Unsafe apps:** the mobile phone vendors try to ensure app security through requiring apps to be signed to be downloaded from official app stores.

6. **Lost Devices:** Portable devices are easily lost or stolen.

5.1.3 Installation of Antivirus software

Antivirus software is a class of program designed to prevent, detect and remove malware infections on individual computing devices, networks and IT systems. There are many antivirus products available in market like Avast, McAfee, Kaspersky, Norton, Trend, AVG Anti Virus, AVIRA, PANDA etc. Now we are considering Avast antivirus and below are the steps to install and use this product.

There are many antivirus software are available in market like Avast, Mcfree, 360Total Security, Kaspersky, AVG etc.

Here we are considering 360TotalSecurity antivirus for installation.

1. Open the web browser and download free 360TotalSecurity antivirus.
2. Locate your program under the “Products” tab on the 360 Total Security website. Click the “Download” button to download the anti-virus software for free.
3. Once the program downloads, locate the installer file and double click to open.
360 tal Security: 360TS_Setup_Mini
4. Select your language option and click “Install” to continue the installation process.
5. After installation click on start to setup antivirus.
6. Using antivirus software can perform operation like virus scan, clean-up, speed up etc.